



Designing Artificial Intelligence Products

Week 1 Office Hour
31 Aug 2026

Danny Malter



Introduction and Instructions

Questions

Be very specific when asking a question

A good question would be something like:

- “On Assignment 2.1 Question 3 ...”
- “In module 3, video 5...”
- If necessary, include screenshots



Discussion – Costs around ML

1. GPU training costs

- [AWS EC2 on-demand pricing](#)
- [Azure Virtual Machines pricing](#)
- [Google Cloud pricing calculator](#)

2. Training data generation costs

- Cost of data
- Cost of employees
- Cost of compute resources

3. Error & bias removal costs

- Identifying and correcting model errors
- Detecting and mitigating bias in outputs

4. Per-query pricing in production

- ChatGPT: a few cents per query (OpenAI)
- Google Places API: about \$0.02 per query — [pricing](#)

The First Stage of the AI Design Process

Stage 1 — Intelligence

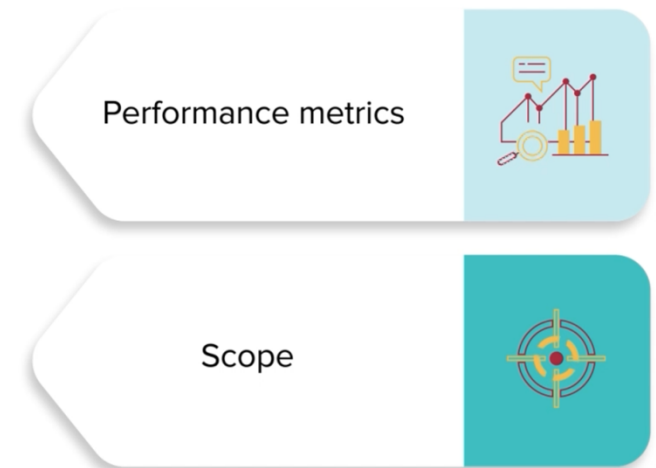
The first stage defines the system's core capability: what it can do, and how well it does it. It is assessed through two lenses.

- **Performance metrics** — how quality is measured: accuracy, efficiency, robustness, fairness, and usability.
- **Scope** — the exact problem it solves and the boundaries of what it should and shouldn't handle.



Intelligence

Stage one



Intelligence: Performance Metrics



Predictive accuracy — How close are predictions to reality? (Accuracy, precision, recall, F1, RMSE)



Efficiency — How fast or resource-hungry is the system? (Latency, memory, energy use)



Robustness — How does performance hold up with noisy, missing, or adversarial data?



Fairness & ethics — Do outcomes differ systematically across groups? Are results interpretable?



Usability — Does the system meet real user needs in context, not just in testing?

Intelligence: Scope



Problem Framing

What exact question are you answering, and why does it matter? Is it classification, prediction, optimization, or generative?



Use Case Boundaries

What should the AI handle, and what is explicitly out of scope? (e.g., fraud detection for credit cards, not loans).



Granularity

How broad should the model's capabilities be? A focused, narrow scope usually outperforms a broad, vague one.



Stakeholder Expectations

Who will use the system, and what outcomes do they care about? Align scope with user needs and org goals.



Data Fit

Does the available data actually support the scoped problem? If not, should you narrow or adjust the scope?



Scalability

Think ahead: if the model succeeds, can the scope expand without breaking the system's design?

The Second Stage of the AI Design Process

Stage 2 — Business Process

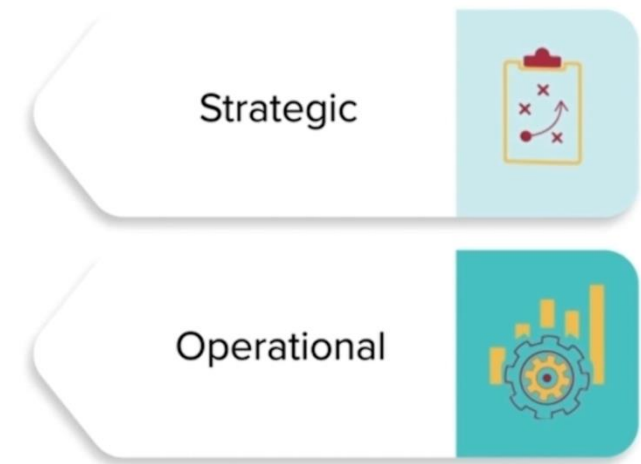
The second stage aligns AI with how the business creates value, seen from two angles.

- **Strategic** — where AI can strengthen the company's competitive position.
- **Operational** — where AI can improve day-to-day processes and efficiency.



Business process

Stage two

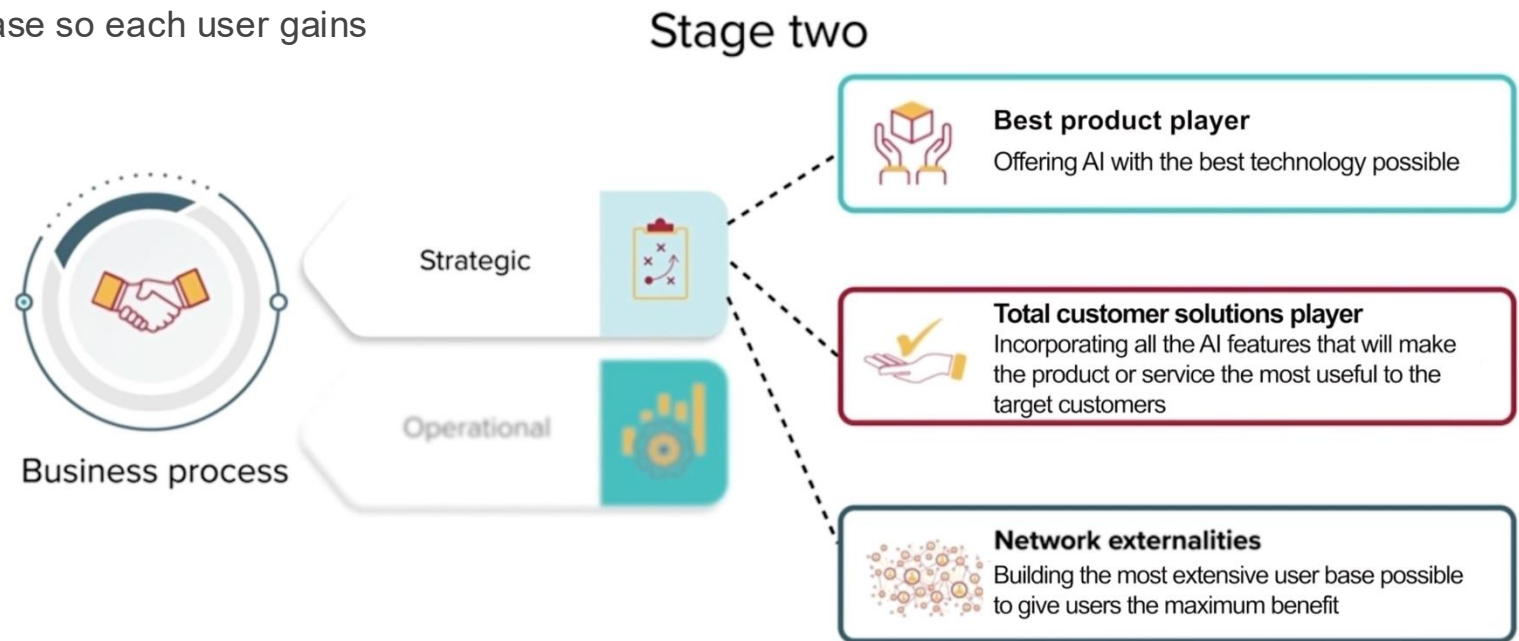


The Second Stage of the AI Design Process

Stage 2 — Strategic

Strategically, AI can position the business to win in three ways.

- **Best product player** — offer AI built on the best technology possible.
- **Total customer solutions player** — combine AI features to make the product most useful to target customers.
- **Network externalities** — build the widest user base so each user gains more value.

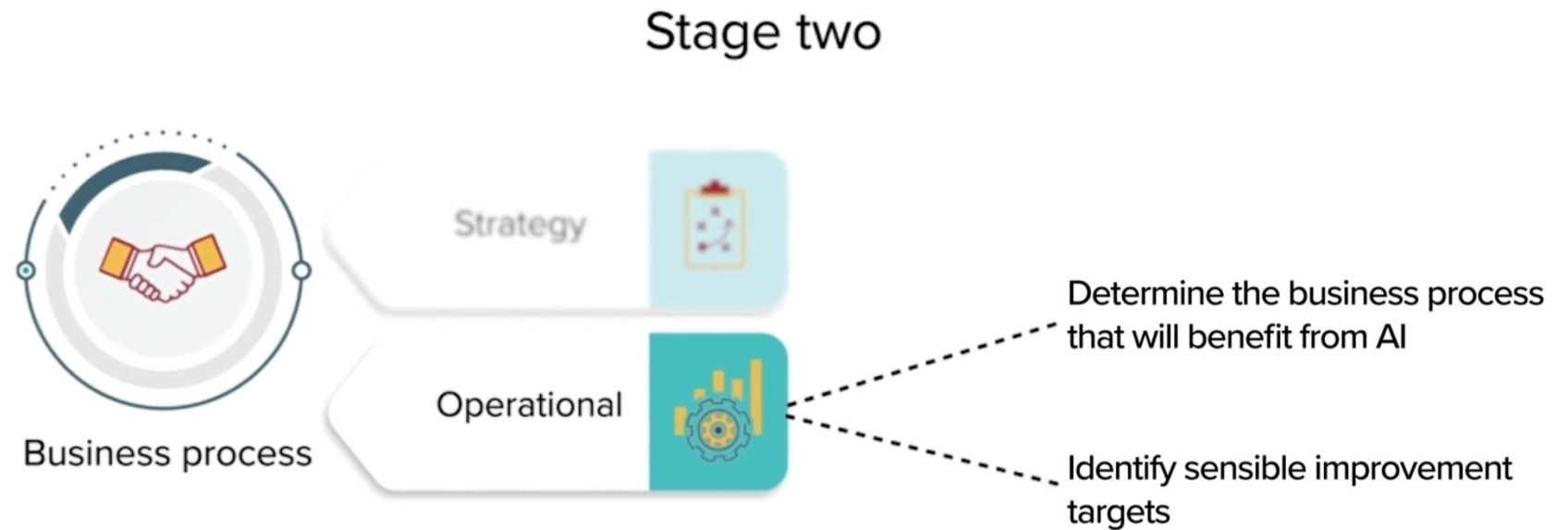


The Second Stage of the AI Design Process

Stage 2 — Operational

Operationally, focus AI where it delivers measurable process gains.

- **Determine the business process** — pinpoint the process that will benefit most from AI.
- **Set improvement targets** — aim for measurable gains — improve speed and improve accuracy.



The Third Stage of the AI Design Process

Stage 3 — AI Technology

The third stage builds the technology and data foundation, across two areas.

- **Intellectual property (IP)** — protect the models, methods, and innovations you create.
- **Data approach** — assess what data is available, what type you have, and where it is located.



AI technology

Stage three

Intellectual
property
(IP)



Data
approach
(FAANGs)



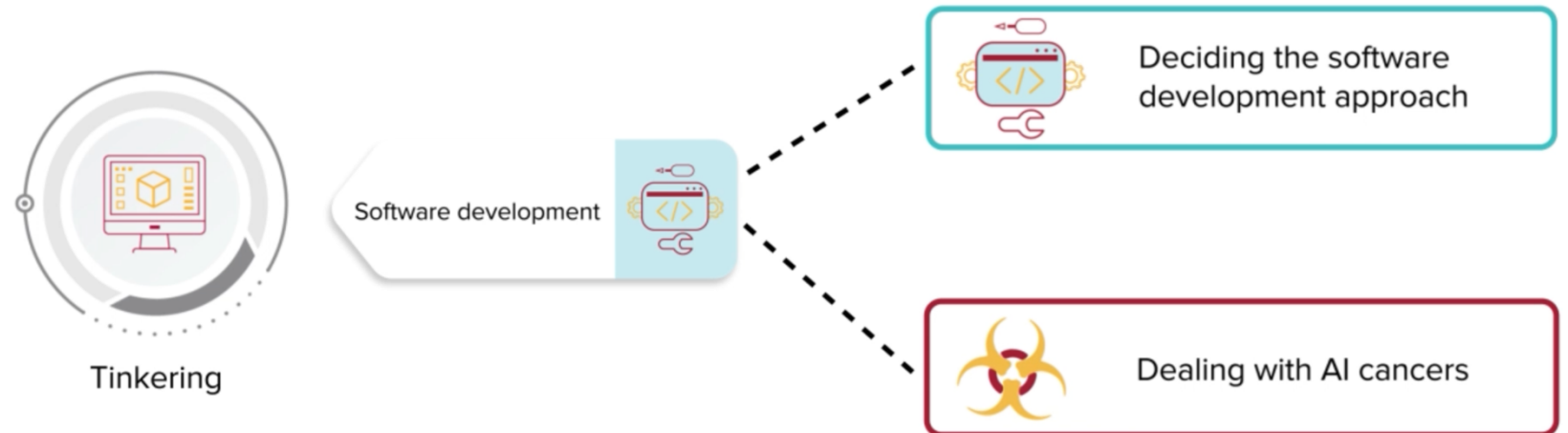
The Fourth Stage of the AI Design Process

Stage 4 — Software Development

The fourth stage is about building and maintaining the system responsibly.

- **Development approach** — decide how the AI system will be built, trained, and iterated.
- **Guard against “AI cancers”** — prevent unaligned growth and resource drain as the system scales.

Stage four

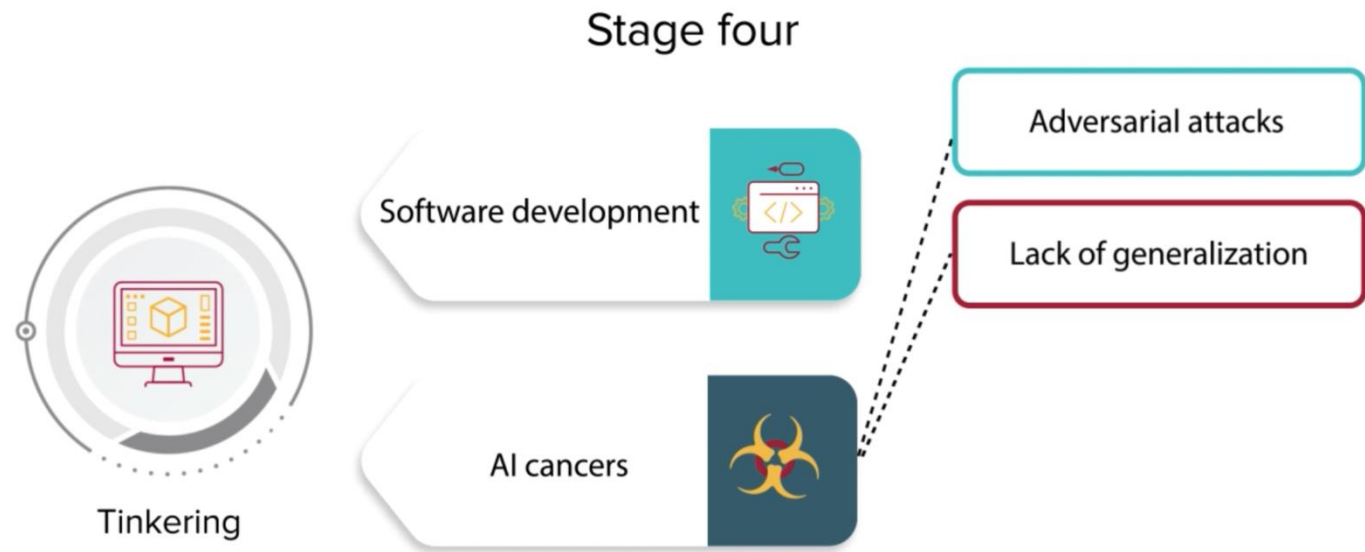


The Fourth Stage of the AI Design Process

Stage 4 — Development Risks

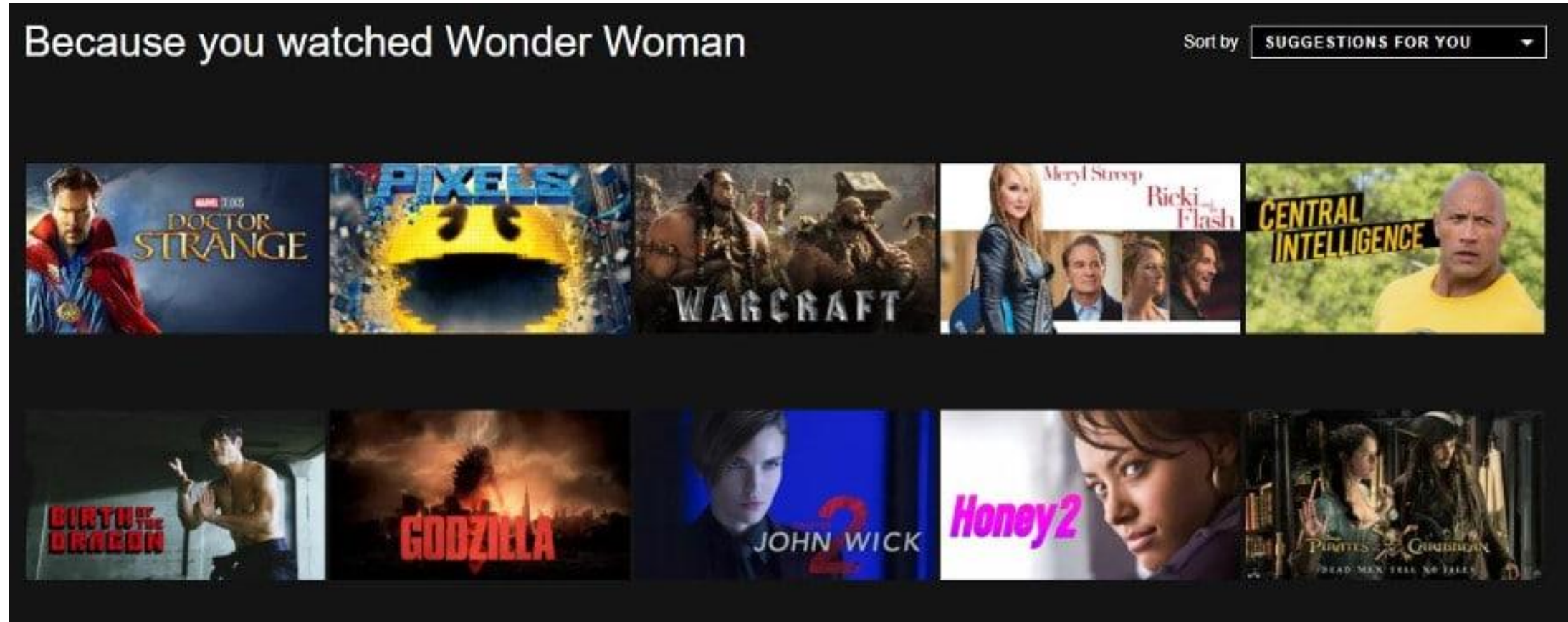
Building the system means anticipating where it can fail or degrade.

- **Software development** — watch for adversarial attacks and lack of generalization.
- **AI cancers** — unaligned growth and resource drain that erode the system over time.



Intro to Machine Learning

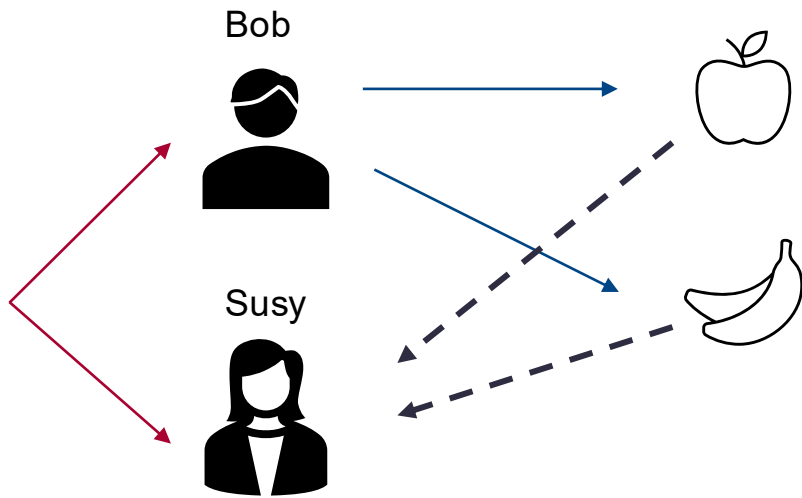
Generating User Content



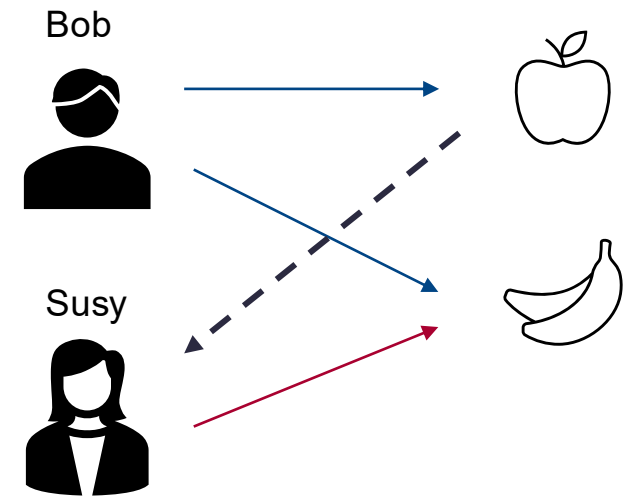
- Cold start problem
- How do we make sure users see new perspectives?

Generating User Content

User Based Recommendations



Item Based Recommendations



Potential Tools for AI Applications

- Great for rapid prototyping and full-stack commercial applications.
- **Modern**, elegant, open-source language.
- Highly **expressive**, i.e., you can be more **productive**.
- Well documented and has an established and **growing community**.
- Comes with **“batteries included”** - in other words, both languages have libraries that will help you do a ton of different tasks!

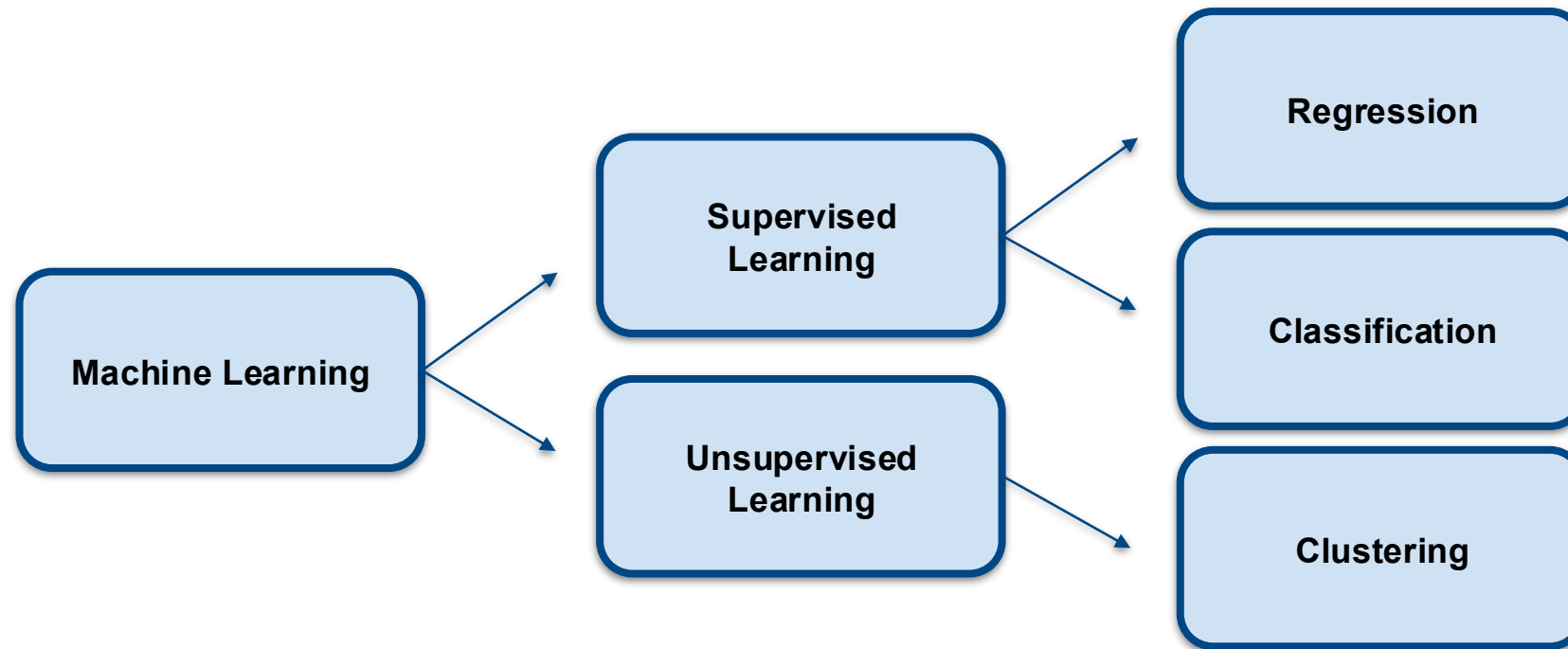


Machine Learning

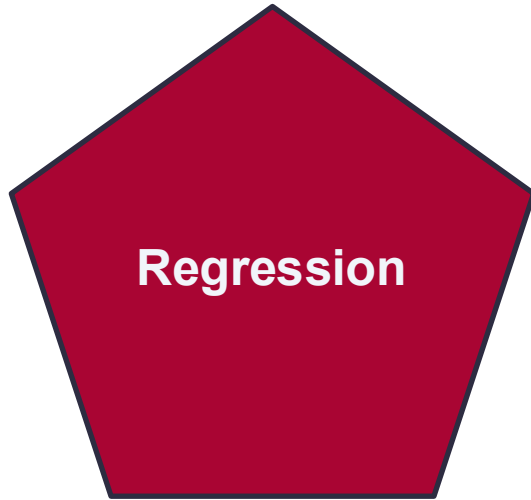
Machine learning is a **branch of artificial intelligence**. It is concerned with the construction and study of systems that can learn from data.

Supervised learning: *Making a prediction*

Unsupervised learning: *Inferring a function to describe hidden structure from “unlabeled” data*



Supervised Machine Learning

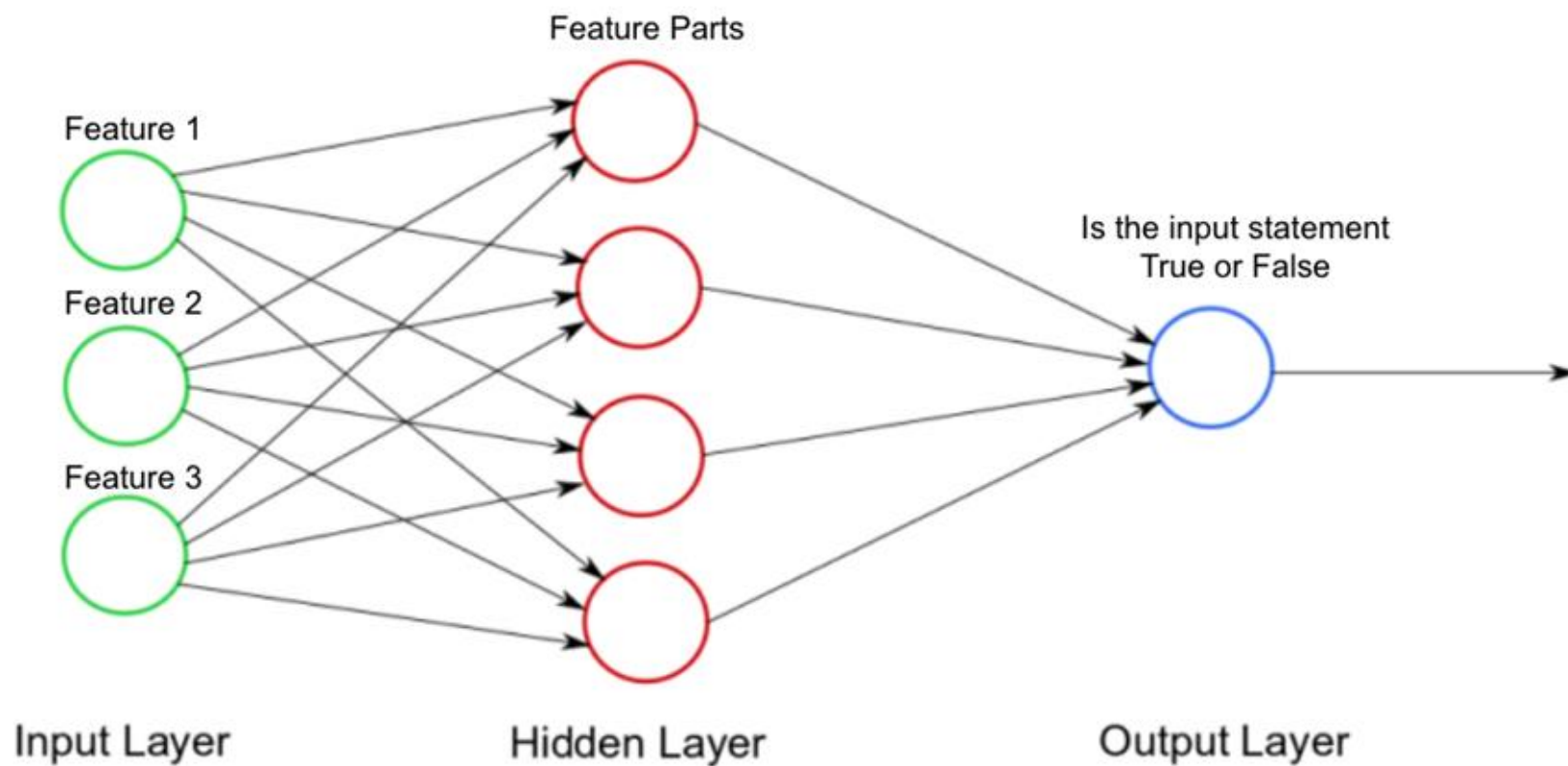


Predicting home prices in a
given neighborhood

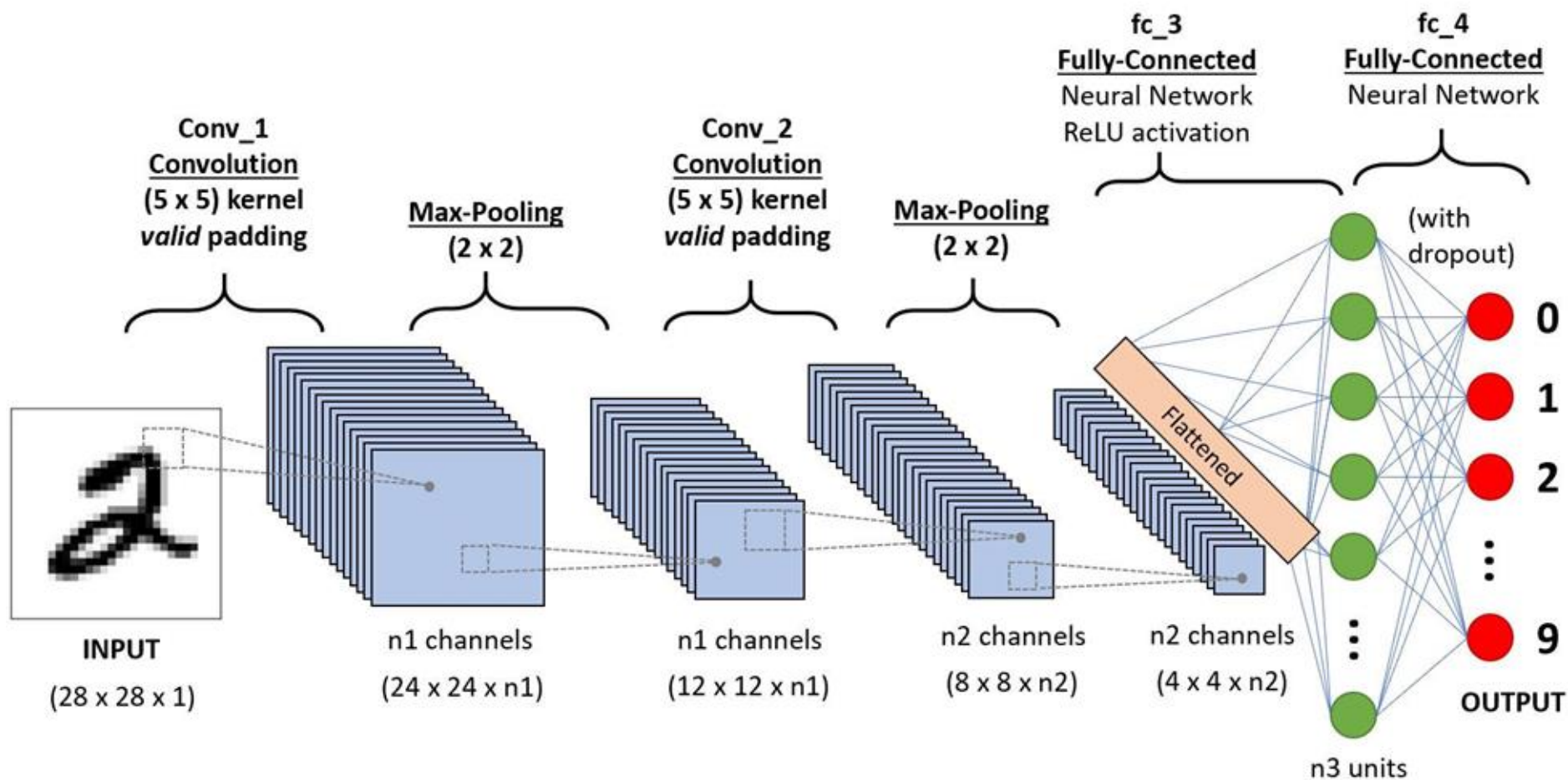
Classifying a tumor as
benign or malignant



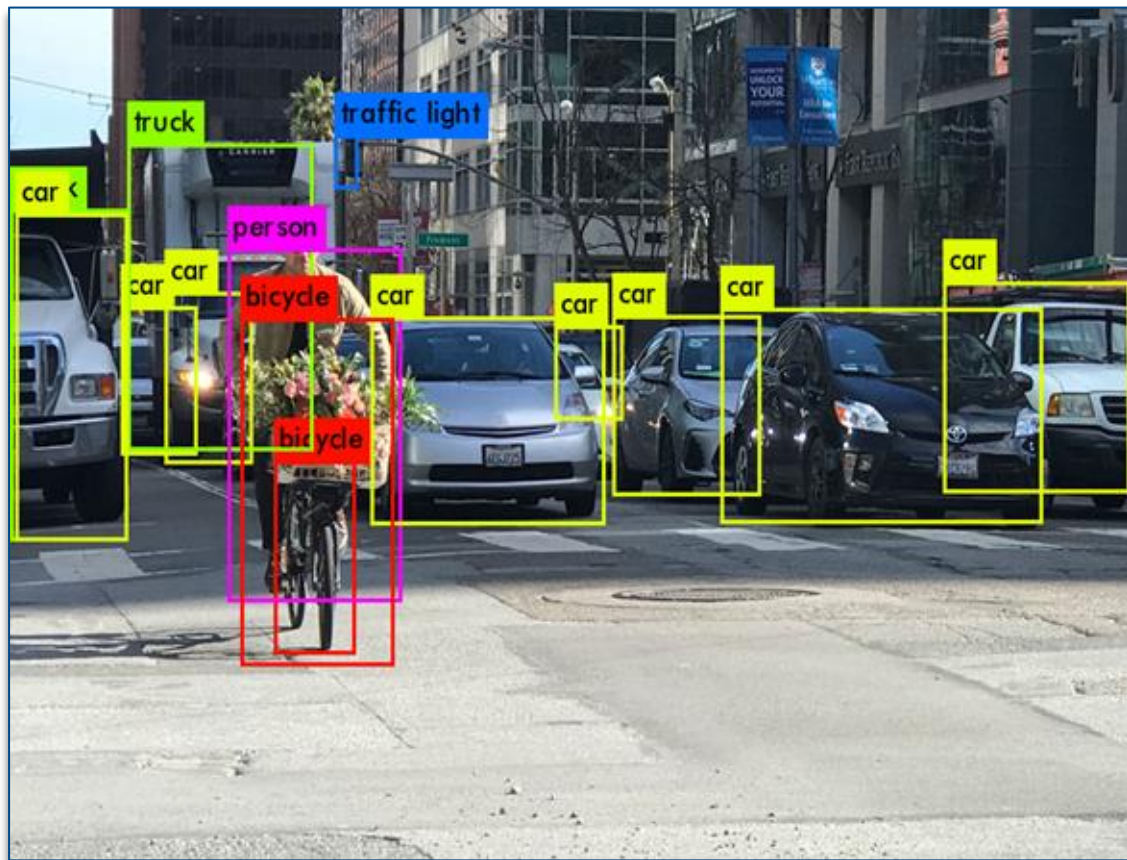
Supervised Machine Learning: Neural Networks



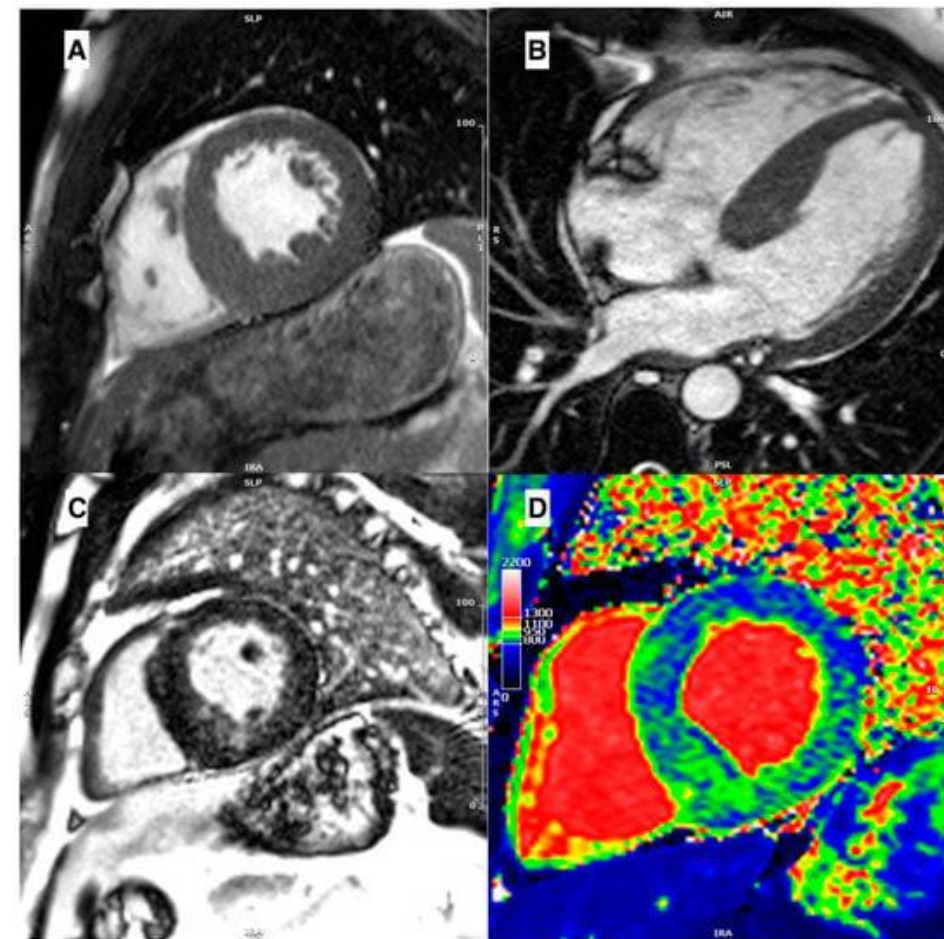
Supervised Machine Learning: Convolutional Neural Networks



Supervised Machine Learning: Deep Learning



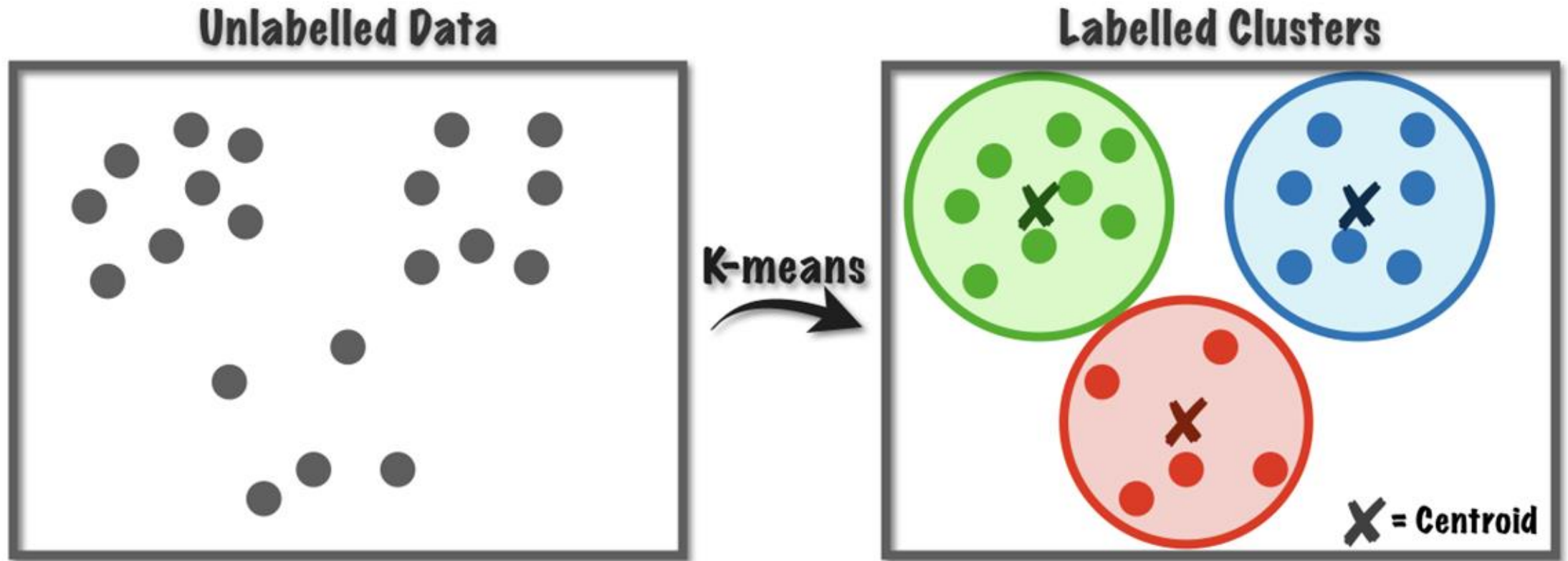
Sources: Deep Learning YOLO



Unsupervised Machine Learning

- Suppose you want to understand your customer base so that you can produce appropriate segments that you can target with your next marketing campaign.
- You have a set of data about your customers, including age, location, previous purchases, time of visit, etc.
- Types of unsupervised learning algorithms:
 - **Clustering**
 - K-means
 - Hierarchical
 - **Anomaly detection**

Unsupervised Machine Learning: K-Means Clustering



Unsupervised Machine Learning: Anomaly Detection

